

UEC612: DIGITAL SYSTEM DESIGN

L	T	P	Cr
3	1	2	4.5

Course Objective: To familiarize the student with the analysis, design and evaluation of digital systems of medium complexity that are based on SSI, MSI and Programmable logic devices. Also, to familiarize the students with the issues in the design of iterative networks, timing analysis of synchronous and asynchronous systems.

Syllabus

Binary Codes: Review of special binary codes, Error detection and correction codes.

Combinational Circuits: Q. M. Method, Variable Map Method, Ripple carry adder, BCD adder, High speed adder, Subtractor, Code conversion, Magnitude comparators, Applications of Encoders, Decoders, MUX, DEMUX, Implementations using ROM, PLA, PAL. Standard ICs and their applications. Using combinational modules to design digital systems, Iterative networks.

Sequential Circuits: Various types of latches and flip-flops and their conversions, Universal Shift Registers, Counters – Ring, Johnson, Design of Counters, Timing issues, Setup and hold times, operating frequency limitations, Static Timing Analysis, Standard ICs for their applications, Finite State Machines – Moore and Mealy, Design of Synchronous and Asynchronous sequential circuits, Races and hazards, hazard free design.

Logic Circuits: DTL, TTL, MOS, CMOS logic families their comparison, Detailed study of TTL & CMOS logic families and their characteristics i.e. Fan-in, Fan-out, Unit load, Propagation delay, Power dissipation, Current & voltage parameters, Tristate Logic, Interfacing of TTL & CMOS logic families, reading and analyzing Datasheets, Performance estimation of digital systems.

HDL: Introduction, Structure of HDL Module, Operators, Data types, Types of Descriptions, Simulation and synthesis, Brief comparison of VHDL and Verilog. Data-Flow Descriptions: Highlights of Data flow descriptions, Structure of data-flow description, Data type-vectors.

Familiarization with standards: IEEE 91a-1991 and IEEE 91-1984.

Laboratory Work: To study standard ICs and their usage, latches and Flip-flops, Design of registers and asynchronous/synchronous up/down counters, Variable modulus counters, Design of Finite State Machines, Study of timing waveforms, Usage of IC tester. Programming of combinational and sequential systems using VHDL or Verilog.

Course Learning Objectives (CLO)

The students will be able to:

1. Apply various minimization methods to simplify the boolean functions and construct the circuit using logic gates
2. Design the combinational circuits and iterative networks using basic modules.
3. Design and analyze sequential circuits and systems.
4. Evaluate the performance of ICs of the logic families.
5. Design, simulate and synthesize the digital systems using HDL.

Text Books

1. Fletcher, W.I., Engineering Approach to Digital Design, Prentice Hall of India 4th ed., 2007
2. Wakerly, J.F., Digital Design Principles and Practices, Prentice Hall of India, 5th ed., 2013

Reference Books

1. Givone D. D., Digital Principles and Design, Tata McGraw Hill (2007) 2nd ed.
2. Tocci, R.J., Digital Systems: Principles and Applications, Prentice-Hall (2006) 10th ed.
3. Wakerly, J.F., Digital Design Principles and Practices, Prentice Hall of India (2013) 5th ed.
4. Bhaskar J., A VHDL Primer, Prentice Hall of India (1999) 3rd ed.
5. Palnitkar S., Verilog HDL - A Guide to Digital Design and Synthesis, (2003) 2nd ed.